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Assignment - II

① Explain Expectation value, Eigen function & Eigen values.

-> Expectation value :- In Quantum mechanics, the expectation value is the probabilistic expected value of the result (measurement) of an experiment, can be thought of an average of all the possible outcomes of a measurement.

- Expectation value is as such is not the next most probable value of the measurement.
- Explanation :- The result of measurement of the position x is a continuous random variable. Consider a wave function $\psi(x)$: The $|\psi|^2$ value is the probability density for the position and $|\psi|^2 dx$ is the probability of finding the particle between x and $x + dx$. If the measurement is repeated manytimes, many possible outcomes are possible and the expectation value of these outcomes can be expressed as

$$\langle x \rangle = \int_{-\infty}^{\infty} |\psi|^2 dx$$

-> Eigen function :-

- Eigen function are those wave functions in quantum mechanics which posses the following properties:

1. They are single valued.
2. Finite everywhere
3. The wave functions and their first derivatives with respect to their variables are continuous.

Mathematical formula = $L[F(x)] = \lambda f(x)$

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→ Eigen values:

• If the wave function is operated by a quantum mechanical operator such that we get back the wave function back multiplied by some constant. is called as Eigen value.

$\hat{H}(\psi) = \lambda(\psi)$

where λ = constant → is called as eigen value

If the quantum mechanical operator is Energy operator, the λ is termed as energy eigen value.

2. Assuming the time independent Schrondinger wave equation discuss the solution for a particle in one-dimensional potential well of infinite height. Hence obtain the normalized wave function.

→ The Time independent Schrondinger wave equation consider a particle of mass 'm' moving with veloci The de-broglie wavelength 'λ' is

$$\lambda = \frac{h}{mv} = \frac{h}{p} \quad -(1)$$ where 'mv' is the momentum of the particle

The wave eqⁿ is

$$\psi = Ae^{i(kx-\omega t)} \quad -(2)$$

where 'A' is a constant and 'ω' is the angular freque of the wave.

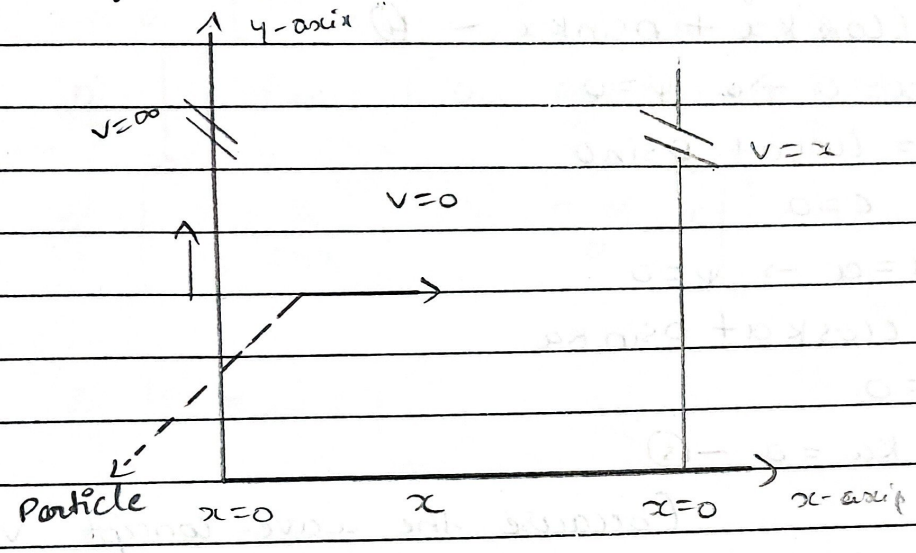
Differenting eq (2) with respect to 't' twice

$$\frac{d^2\psi}{dt^2} = -A\omega^2 e^{(ikx-\omega t)} = -\omega^2\psi \quad -(3)$$

The equation of a travelling wave is

This is time independent Schrondinger wave equation for one dimensional case.

-> Eigen value Energy of a particle in one dimensional infinite potential well (potential well of infinite depth) or of a particle in a box.



Consider a particle of a mass 'm' free to move in dimension along positive x-direction blw x=0 to x=a. the potential energy outside this region is infinite and within the region is zero. the particle is bound state. Such a configuration of potential in space is called infinite potential well. It is also called particle in a box. The schrondinger equation outside the well is

$$\frac{d^2\psi}{dx^2} + \frac{8\pi^2 m}{h^2}(E-\infty)\psi = 0 \quad - (1) \quad \therefore V = \infty$$

for outside, the equation holds good if $\psi = 0$ & $|\psi|$ that is particle cannot be found outside the well and also at the walls.

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The Schrondinger's equation inside the well is:

$$\frac{d^2\psi}{dx^2} + \frac{8\pi^2}{h^2} m E\psi = 0 \quad -② \qquad \therefore V=0$$

let $\frac{8\pi^2 m}{h^2} E = k^2$ $-③$

$$\frac{d^2\psi}{dx^2} + k^2\psi = 0$$

The solution of above equation is;

$$\psi = C\cos kx + D\sin kx \quad -④$$

at $x = 0 \rightarrow \psi = 0$

$$0 = C\cos 0 + D\sin 0$$

$$\therefore C = 0$$

Also $x = a \rightarrow \psi = 0$

$$0 = C\cos ka + D\sin ka$$

But $C = 0$

$$\therefore D\sin ka = 0 \quad -⑤$$

$D \neq 0$ (because the wave concept vanishes

i.e $ka = n\pi$ where $n = 0, 1, 2, 3, 4\ldots$ (quantum number)

$$k = \frac{n\pi}{a}$$

Sub eqn (5) and (6) in (4)

$$\psi_n = D\sin\frac{n\pi}{a}x \rightarrow ⑦$$

This gives permitted wave functions

The energy Eigen value function given by

Substitute eqn ⑥ in ③

$$\frac{8\pi^2 m}{h^2} E = k^2 = \frac{n^2\pi^2}{a^2}$$

$$E = \frac{n^2 h^2}{8ma^2}$$

this is the expression for energy Eigen value.

To find out the value of D normalization of the wave function it to be done.

i.e $\int_0^a |\psi_n^2| \, dx = 1 \quad -(8)$

using the wave of ψ_n from eqn: (7)

$$\int_0^a D^2 \sin^2 \frac{n\pi}{a} x \, dx = 1$$

$$\frac{D2}{} \int_0^a \left[1 - \cos \left(2n\pi / (a) x \right) \right] dx = 1$$

$$\because \sin^2\theta = \left[\frac{1 - \cos^2\theta}{2} \right]$$

$$\frac{D^2}{2} \left[\int_0^a dx - \int_0^a \cos \frac{2n\pi}{a} x \, dx \right] = 1$$

$$\frac{D^2}{2} \left[x - \frac{a}{2n\pi} - \sin \frac{2n\pi}{a} x \right]_0^a = 1$$

$$\frac{D^2}{2} [a = 0] = 1$$

$$\frac{D^2}{2} a = 1$$

$$D = \sqrt{\frac{2}{a}}$$

the normalized wave functions of a particle in one dimensional infinite potential well is :

$$\psi_n = \sqrt{\frac{2}{a}} \sin \frac{n\pi}{a} x \quad -(a)$$

③ Discuss the wave functions, probability densities and energy eigen values for a particle in a box with diagram. [Assume normalized wave function of the particle.]

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Let us consider the most probable location of the particle in the will and its energies for first three cases.

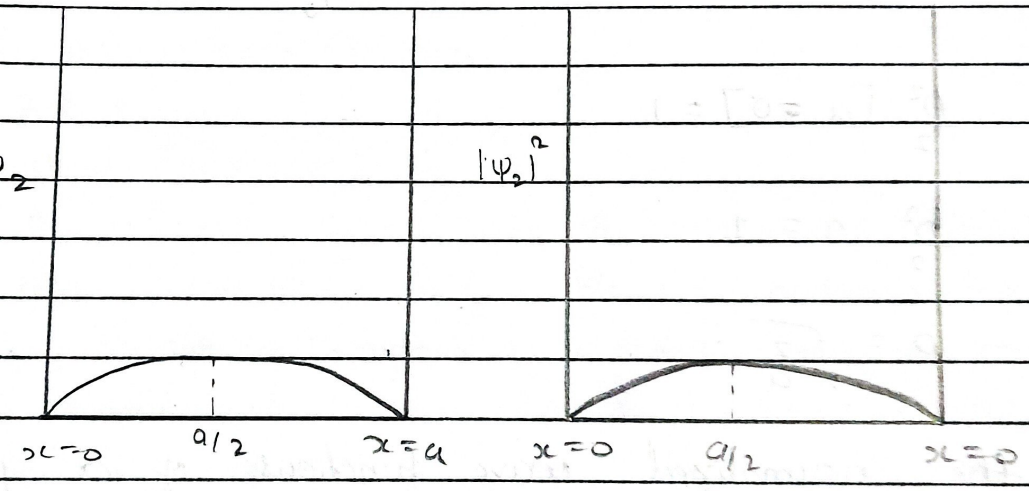
Case I → $n=1$ It is the ground state and the particle is normally present in this state

The Eigen function is

$$\psi_1 = \sqrt{\frac{2}{a}} \sin \frac{\pi}{a} x \quad \because \text{from eqn } (7)$$

$\psi_1 = 0$ for $x=0$ and $x=a$

But ψ_1 is maximum when $x = a/2$



The plots of ψ_1 verses x and $|\psi_1|^2$ verses x are shown in the above figure.

$|\psi|^2 = 0$ for $x=0$ and $x=a$ and it is maximum for $x = a/2$. i.e in ground state the particle cannot be found at the walls, but the probability of finding it is maximum in the middle.

The energy of the particle at the ground state is $E_1 = \frac{h^2}{8ma^2} = E_0$

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case II → n = 2

In the first excited state the Eigen function of this state is

$$\psi_2 = \sqrt{\frac{2}{a}} \sin \frac{2\pi}{a} x$$

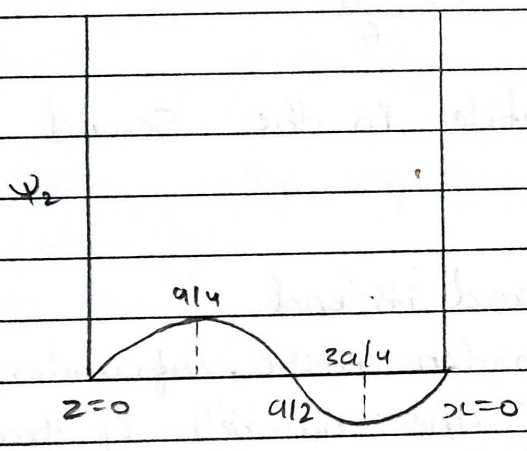
$\psi_2 = 0$ for the value $x = 0, a/2, a$

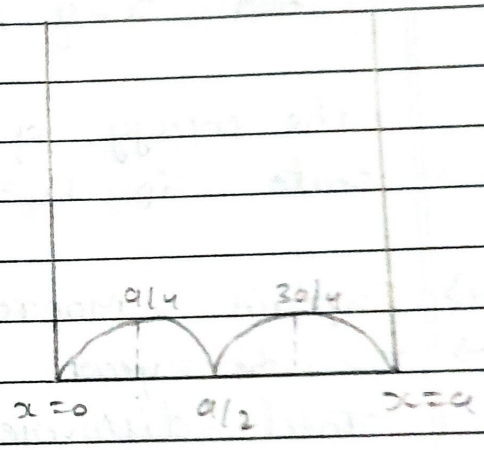
Also ψ_2 is max for the values $x = a/4$ & $3a/4$

these are repeated in the graphs

$|\psi_0|^2 = 0$ at $x = 0, a/2, a$ i.e particle cannot be found either at the centre $|\psi_2|^2$ = maximum

for $x = \frac{a}{4}$, $x = \frac{3a}{4}$





The Energy of the particle in the excited state is $E_2 = 4E_0$

Case III → n = 3

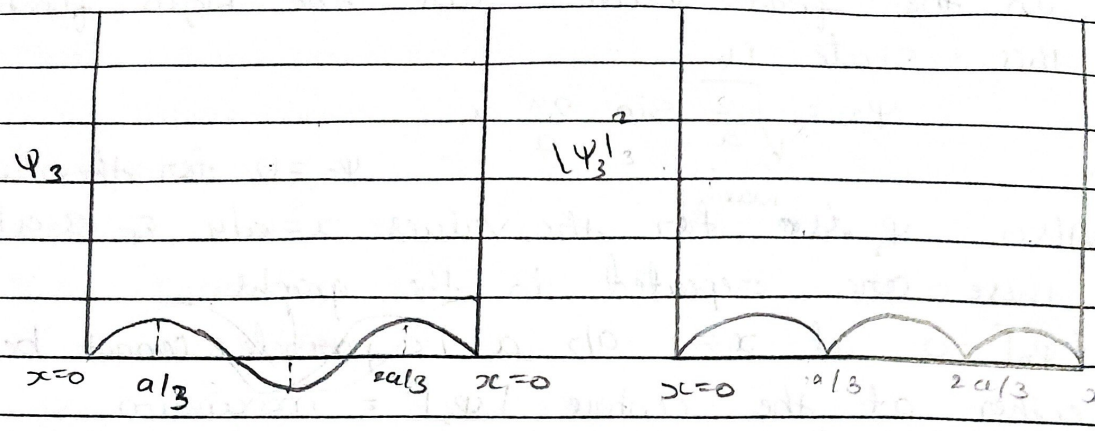
In the second excited state,

$$\psi_2 = \sqrt{\frac{2}{a}} \sin \frac{3\pi}{a} x$$

$\psi_3 = 0$, for $x = 0, a/3, 2a/3$ and a

ψ_3 is maximum for $x = a/6, a/2, 5a/6$

These are represented in the graphs.



$|\psi_3|^2 = 0$ for $x = 0,\ a/3,\ 2a/3$ and a. $|\psi_3|^2$ = maximum for $x = \frac{a}{6},\ x = \frac{a}{2},\ x = \frac{5a}{6}$

The energy of the particle in the second excited state is $E_3 = 9E_0$

4) Explain Moore's law and its end.

→ In the year 1965, Gordon more, cofounder of Intel discovered that "the number of transistors in a dense Integrated circuit (IC) about every two years". Moore's law is an observation and projection of a historical trend. Rather than a law of physics, it is an emphirical relationship linked to gains from experience in production.

→ Moore's law is considerable because it means that computers and their computing power get smaller and faster overtime. though this law is putting the brakes on now and consequently the improvement in classical computers is not like before it used to be. this show that the

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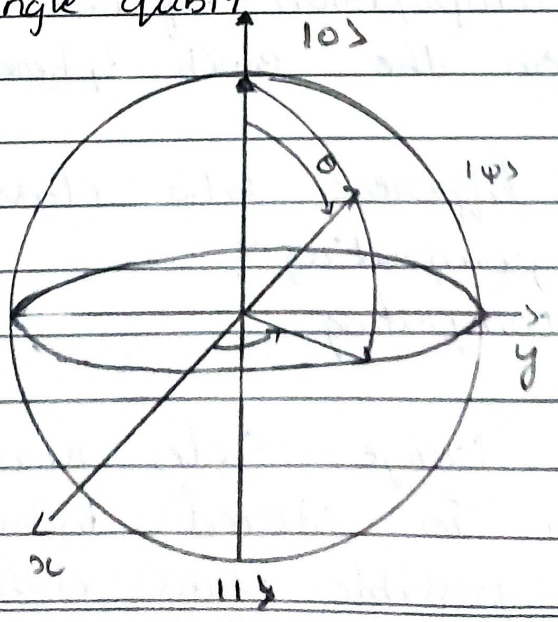
Size of the circuits of the classical computers after 3-2 nanometers has reached their limit. At these scales, controlling the flow of electrons becomes increasingly more difficult as all kinds of quantum effect play themselves with the transipter itself. since, transipter is approaching the point where it is simply as small as we can ever make it and have it still function

-> The way we have been building and improving silicon chips is coming to its final iteration. this quantum computation is the option for the further generation.

5] Explain the representation of quit using Bloch sphere ?

-> The pure state space qubits (Two level quantum mechanical system) can be visualized using an imaginary sphere called Bloch Sphere. It has a unit radius.

- This Bloch sphere picture is elegant and powerful for the single qubit



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→ The Arrow on the sphere represents the state of the Qubit. The north & south poles are used to represent the basis states $|0\rangle$ and $|1\rangle$ respectively. The other locations on the bloch sphere represents the superposition on state i.e $|0\rangle$ and $|1\rangle$ states and represented by $\alpha|0\rangle + \beta|1\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$ thus a qubit can be any point on the bloch spere

- The bloch sphere allows the state of the qubit to be represented in unit spherical co-ordinates they are the polar angle θ and the azimuth angle ϕ. the block sphere is represented by the equation

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle$$

here $0 \leq \theta \leq \pi$ and $0 \leq \phi \leq 2\pi$. The normalization constraint is given by $\left|\cos\frac{\theta}{2}\right|^2 + \left|\sin\frac{\theta}{2}\right|^2 = 1$

- for any gate operation, taking an initial state to final state of the single-qubit is equivalent to a composition of one or more simple notations on the Bloch Sphere.

6) Elucidate difference b/w classical computing and quantum computing?

→ classical computing

1. used by large scale, multipurpose & devices
2. Information is stored in bits. The in discrete number of possible states either 0 or 1.

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3. Power increases in 1:1 relationship with number of transistor.
4. They have low error rates and can operate at room temperature.
5. Calculations are deterministic. This means repeating the same input results in the same output.
6. Data processing is carried out by logical and in sequential order.
7. Operations are governed by Boolean Algebra
8. Circuit behaviour is defined by classical physics

-> Quantum computing:

1. Used by high speed, quantum mechanics based computers
2. Information is based on quantum bits. There is an infinite continuous number of possible states. They are the result of quantum superposition.
3. Power increases in exponentially in proption to number of qubits.
4. They have high error rates and need to be kept at ultra cold.
5. The calculations are probabilistics, meaning there are multiple possible outputs to be the same inputs.
6. Data processing is carried out by quantum logic at parallel instances.
7. Operations are defined by linear algebra by Hibert Space.
8. Circuit behaviour is defined by quantum mechanics.

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4. Explain the matrix representation of 0 and 1 states & apply identity operator to $|0\rangle$ and $|1\rangle$.

-> Matrix Representation of Qubit states $|0\rangle$ and $|1\rangle$

In quantum computing, the basic unit of information is a qubit, which can be represented using vectors (in Dirac notation).

The two standard basis states are:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

These are column vectors in a 2-dimensional complex vector space.

2. Identity operator:

The identity operator is a linear operator that leaves any vector unchanged when applied. In matrix form, the identity operator to $|0\rangle$:-

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

3. Applying Identity Operator to $|0\rangle$ and $|1\rangle$

Let's apply the identity matrix I. to each qubit state:

- Apply I to $|0\rangle$:

$$I|0\rangle = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle$$

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- Apply I to $|1\rangle$:

$$I|1\rangle = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = |1\rangle$$

- The identity operator leaves the qubit state unchanged
- $|0\rangle$ and $|1\rangle$ are represented as vectors in matrix form.
- these form the computational basis in quantum computing.

8. State Pauli Matrices. Apply Pauli Matrices on the states $|0\rangle$ and $|1\rangle$.

→ There are three Pauli matrices, often labeled x, y and z. They are:

1. Pauli - x (Bit-flip gate):

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

2. Pauli - Y (Bit & phase flip gate):

$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

3. Pauli - z (Bit phase - flip gate):

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

- Matrix representation of states $|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

Apply Pauli Matrices to states

1. Pauli-X on $|0\rangle$ and $|1\rangle$

- $X|0\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = |1\rangle$

- $X|1\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle$

Effect: swaps $|0\rangle \leftrightarrow |1\rangle$

2. Pauli-Y on $|0\rangle$ and $|1\rangle$

- $Y|0\rangle = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ i \end{bmatrix} = i|1\rangle$

- $Y|1\rangle = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -i \\ 0 \end{bmatrix} = -i|0\rangle$

Effect: Flips the state with a phase factor:

$Y|0\rangle = i|1\rangle$, $Y|1\rangle = -i|0\rangle$

3. Pauli-Z on $|0\rangle$ and $|1\rangle$

- $Z|0\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle$

- $Z|1\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix} = -|1\rangle$

Effect: Filps the phase of $|1\rangle$:

$Z|0\rangle = |0\rangle$, $Z|1\rangle = -|1\rangle$.

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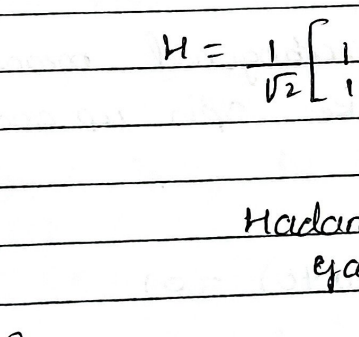
Summary Table

Operator	$\|0\rangle$ Result	$\|1\rangle$ Result
X	$\|1\rangle$	$\|0\rangle$
Y	$i\|1\rangle$	$-i\|0\rangle$
Z	$\|0\rangle$	$-\|1\rangle$

9. Explain Hadamard gate. Apply Hadmard gate on the states $|0\rangle$ and $|0\rangle$

The Hadamard gate is a single qubit gate and is also gate to superposition. H gate acts on single qubit input and produce superposition state output. Matrix form of H gates and its symbol is

$$H = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$



Action of H gate on input.

Let us find out what happens when Hadamard gate operates on a qubit that is the $|0\rangle$ state

$$H|0\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

for $|1\rangle$ state

$$H|1\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}\begin{bmatrix}0\\1\end{bmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\-1\end{bmatrix} = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

$$H|0\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

10. Explain orthogonality and orthonormality with an example for each.

-> orthogonality:-

Two states $|\psi\rangle$ and $|\phi\rangle$ are said to be orthogonal if their inner product is zero mathematically.

$\langle\phi|\psi\rangle = 0$

The two states are orthogonal means they are mutually exclusive. Like spin up and spin down of an electron.

Consider $\langle 0|1\rangle$

$$\langle 0|1\rangle = \begin{bmatrix}1 & 0\end{bmatrix}\begin{bmatrix}0\\1\end{bmatrix} = (0+0) = 0$$

Consider $\langle 1|0\rangle$

$$\langle 1|0\rangle = \begin{bmatrix}0 & 1\end{bmatrix}\begin{bmatrix}1\\0\end{bmatrix} = (0+0) = 0$$

Thus the states $|1\rangle$ and $|0\rangle$ are orthogonal.

-> orthonormality:-

The states $|\psi\rangle$ and $|\phi\rangle$ are said to be orthonormality if

1. $|\psi\rangle$ and $|\phi\rangle$ are normalized

i.e $\langle\psi|\psi\rangle = \langle\phi|\phi\rangle = 1$

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e. $|\psi\rangle$ and $|\phi\rangle$ are orthonormal to each other
ie $\langle\phi|\psi\rangle = 0$

11. Explain the working of phase gate mentioning its matrix representation and its truth table.

-> It is a single qubit gate the phase gate or S gate that transfers $|0\rangle$ into $|0\rangle$ and $|1\rangle$ into $i|1\rangle$. the matrix form of S gate is

$$S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$$

Action of S gate on inputs:

- consider a S gate apply to a state $|0\rangle$ it will remain same

$$S|0\rangle = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle$$

- If S gate apply to a $|1\rangle$ state it will be transformed into $|1\rangle$

$$S|1\rangle = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix} = i\begin{bmatrix} 0 \\ 1 \end{bmatrix} = i|1\rangle$$

- S gate apply to the state $\alpha|0\rangle + \beta|1\rangle$ it transforms to the state $\alpha|0\rangle + i\beta|1\rangle$

$$S|\psi\rangle = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}\begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \alpha \\ i\beta \end{bmatrix}$$

$$S|\psi\rangle = \alpha|0\rangle + i\beta|1\rangle$$

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The S gate representation is as follows:

Input output

The truth table:

Input	output
$\|0\rangle$	$\|0\rangle$
$\|1\rangle$	$i\|1\rangle$
$\alpha\|0\rangle + \beta\|1\rangle$	$\alpha\|0\rangle + i\beta\|1\rangle$

12. Given $A = \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}$, prove that $A^+ = A$

→ Given $A = \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}$

we know that $A^+ = (A^*)^T$

The conjugate matrix A is given by $A^* = \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix}$

Thus $(A^*)^T = \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix}^T = \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix}$

So

$$A^+ = \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}$$

But the original matrix was:

$$A = \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}$$

Thus,

$$A^+ \neq A$$